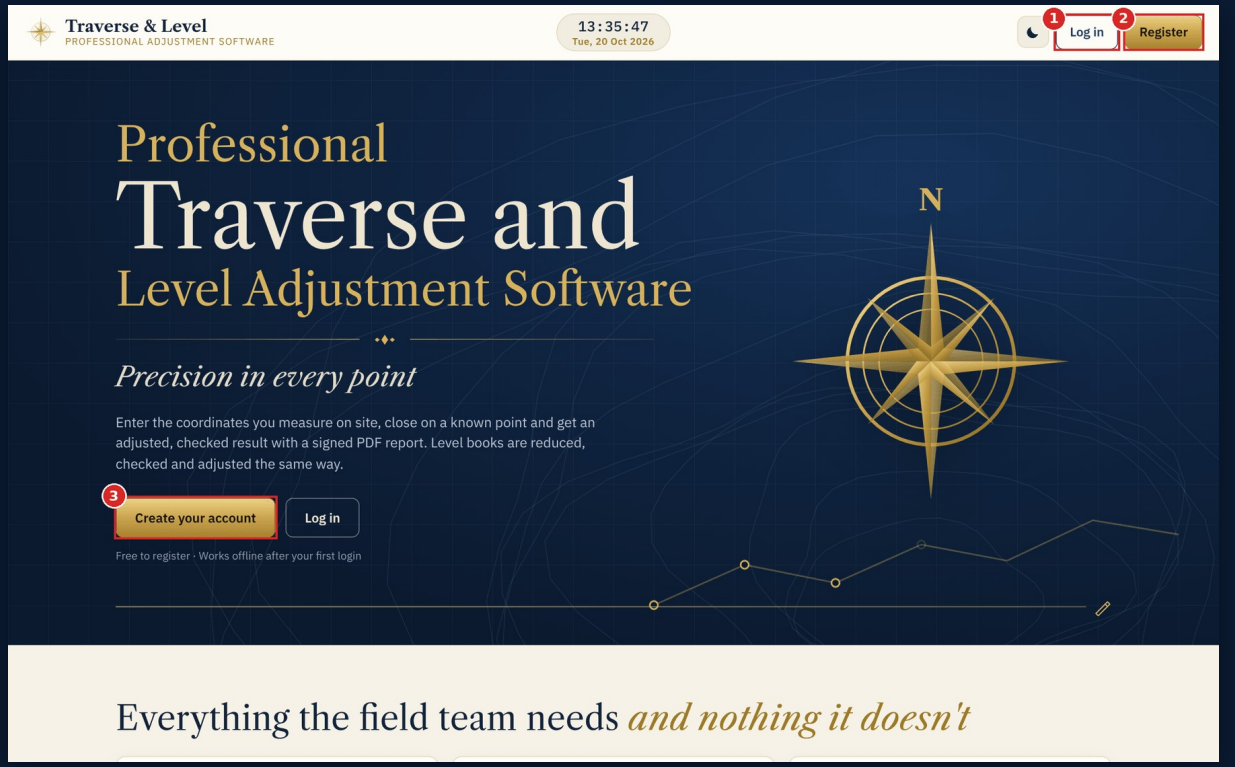
USER MANUAL

Traverse & Level Adjustment

Closed and open traverses, levelling, simulator, saved jobs and PDF reports, explained screen by screen.



Version with open-traverse closing point | October 2026

How to use this manual

Every screenshot carries red numbered boxes. Find the number on the picture, then read the note with the same number beside it.

1

Getting started

Account, top bar and navigation

Slides 4-6

2

Closed traverse

Known points, readings, adjustment, results

Slides 7-15

3

Open traverse (new)

Optional closing point for checking and adjusting

Slides 16-21

4

Levelling

Benchmarks, staff readings, reduced levels

Slides 22-26

5

Simulator, jobs and tips

Practice tool, dashboard, sharing, quick reference

Slides 27-32

TIP Results update as you type, so there is no Calculate button. Fix anything shown in red and the answer appears by itself.

What the software does

Traverse adjustment

Enter known and measured coordinates. The app finds the misclosure, checks the precision you require, and adjusts the points by Bowditch, Transit or Equal distribution.

Levelling

Reduce staff readings by Rise & Fall or Height of Instrument. It checks the arithmetic, compares the misclosure with the allowable value and adjusts the levels.

Traverse simulator

Watch a total station observe a traverse with realistic random errors, then send the result to the Traverse tool to practise adjusting it.

PDF reports and CSV

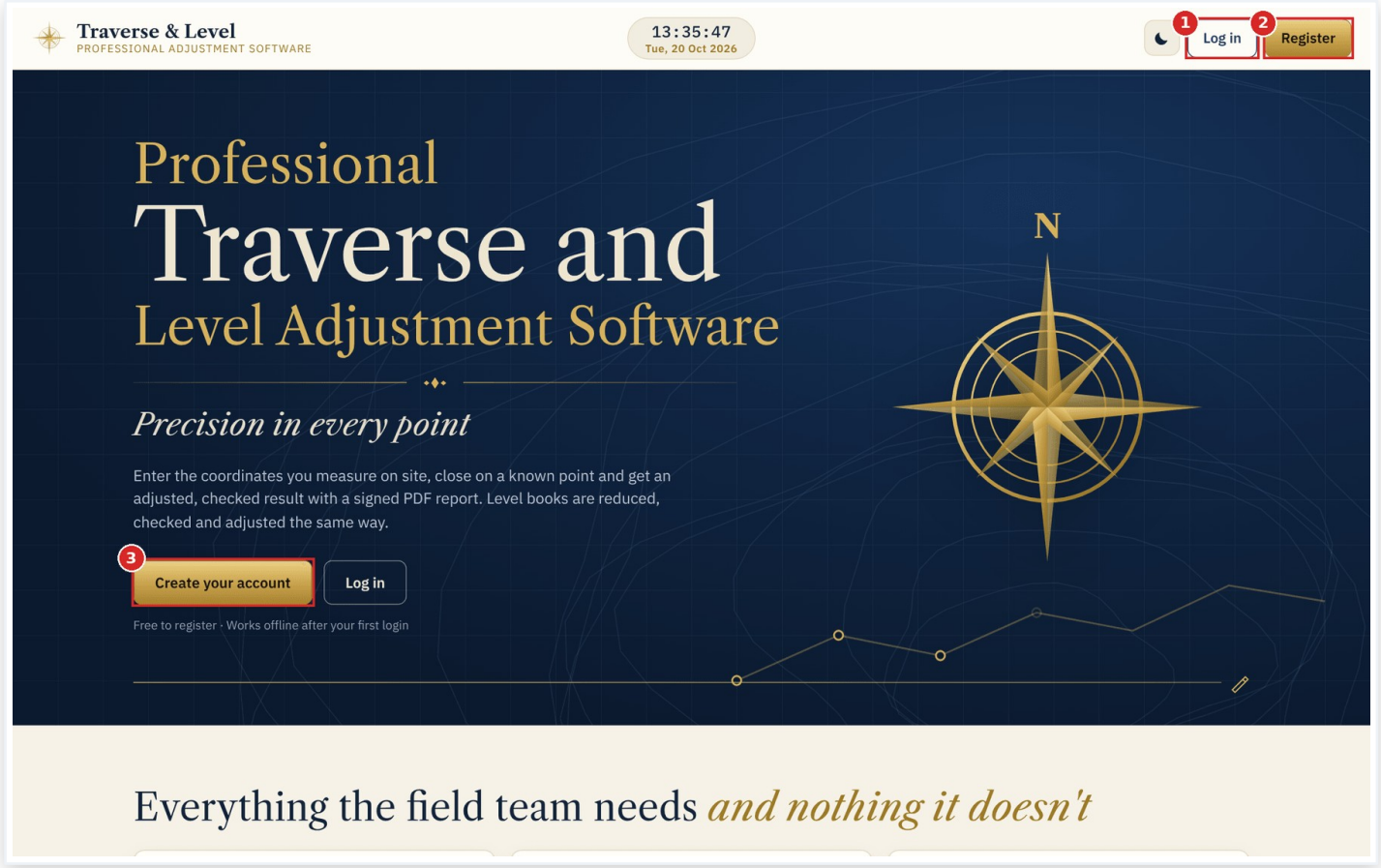
Saved jobs and offline use

Share jobs with colleagues

Light and dark themes

NEW Open traverses can now be checked on a closing point (section 3). Without one they are shown as measured, exactly as before.

The home screen



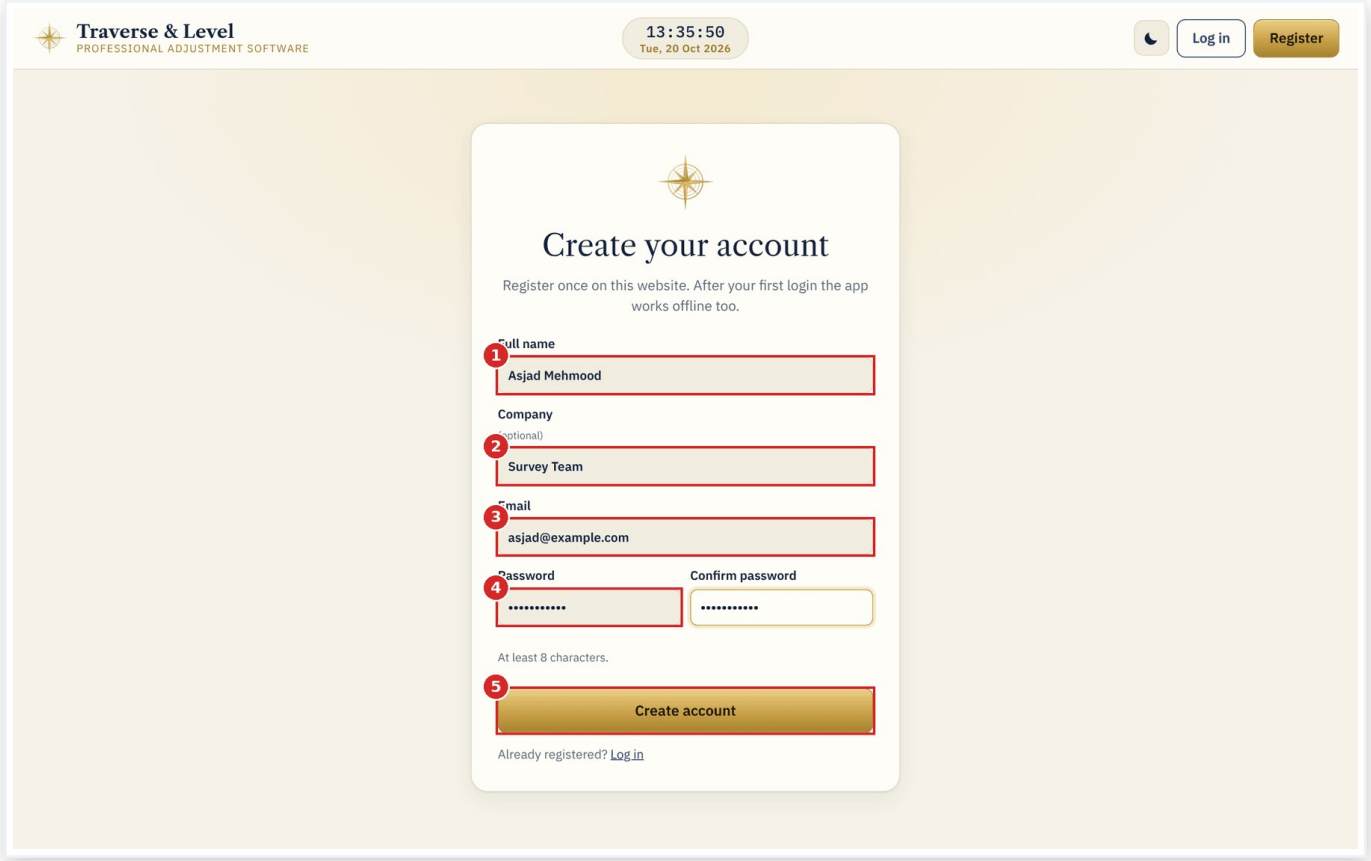
1 Log in
For returning users. Enter your e-mail and password. After the first login the app also works offline.

2 Register
Opens the account form on the next slide. Registration is free.

3 Create your account
The same as Register, as the large button in the banner. Scroll down for a tour of every tool.

NOTE A yellow "Demo mode" bar means the account server is not connected. Accounts and jobs are then kept only on this computer.

Create your account



1. **Full name**
   Shown in the dashboard greeting and used as the Surveyor on reports.

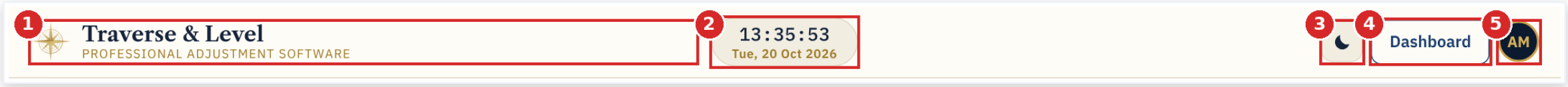
2. **Company (optional)**
   Free text for your organisation.

3. **E-mail**
   Your login name. Colleagues also use it to share jobs with you.

4. **Password and confirm**
   At least 8 characters. Both boxes must match or an error appears.

5. **Create account**
   Takes you to your dashboard with a welcome message.

Top bar and navigation



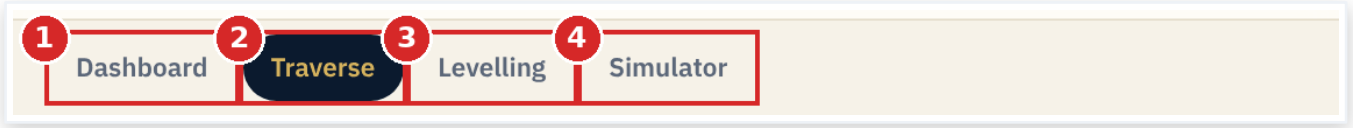
1. **App name**
   Click to return to the home page.

2. **Clock**
   Live date and time. Click it to open a calendar.

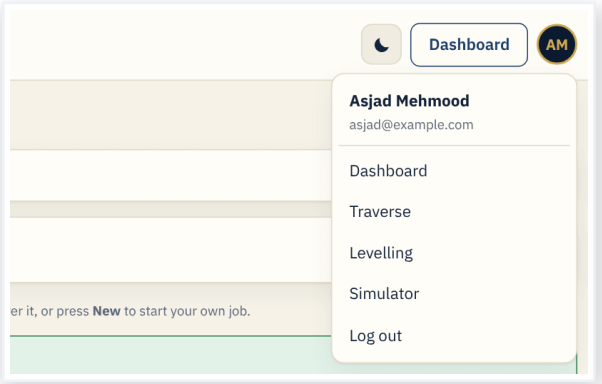
3. **Light / dark**
   Switches the colour theme. Your choice is remembered.

4. **Dashboard**
   Your saved jobs and summary. Visible after login.

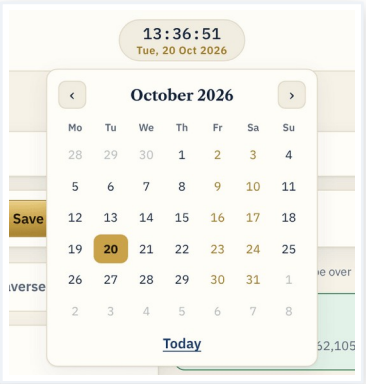
5. **Account button**
   Shows your name and e-mail, links to every tool, and Log out.



Navigation tabs, numbered left to right: Dashboard, Traverse, Levelling, Simulator. The tab you are on is dark.

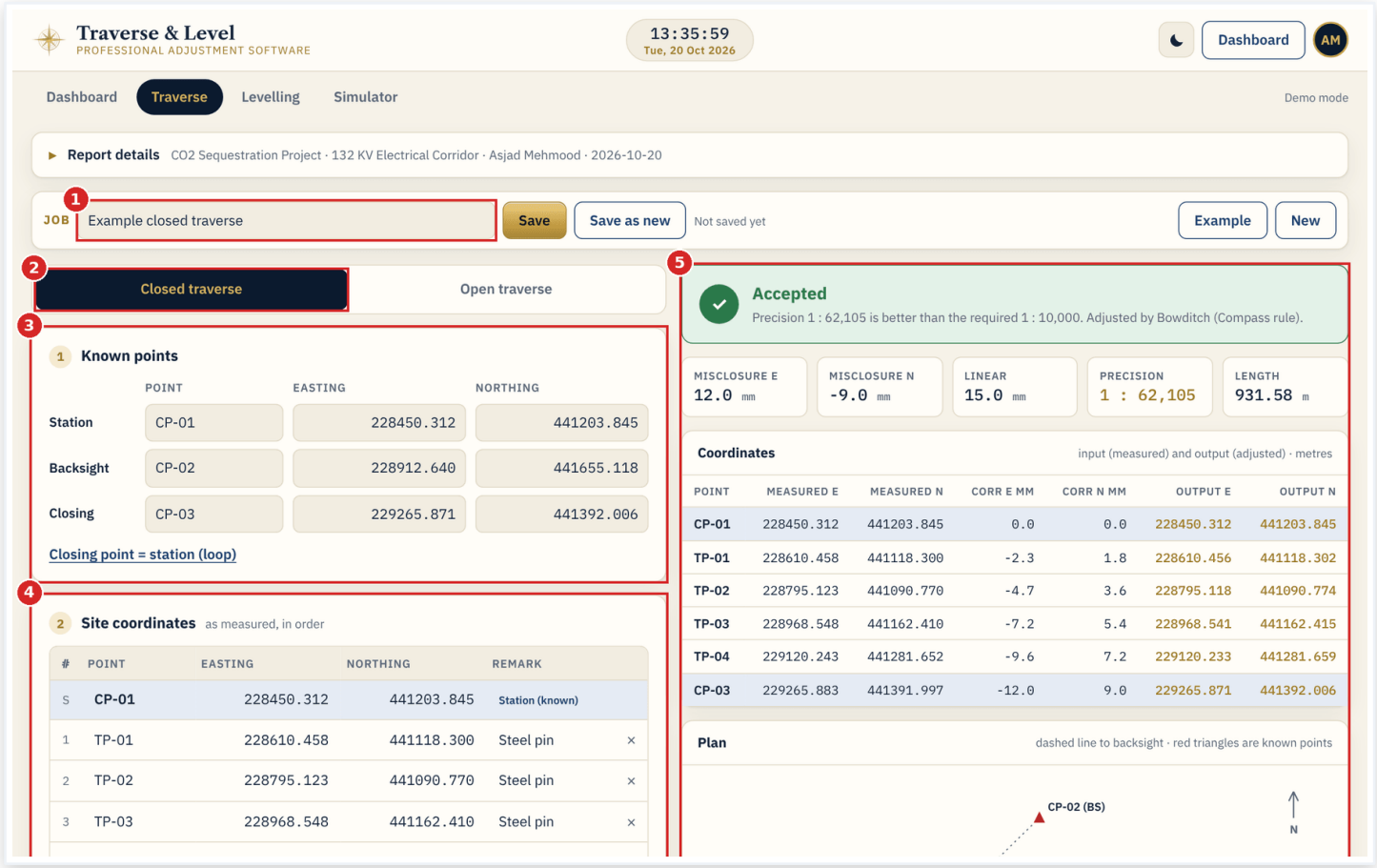


Account menu



Calendar

The Traverse screen



1 Job bar
Name, save, example and new.

2 Traverse type
Closed checks on a known point. Open is in section 3.

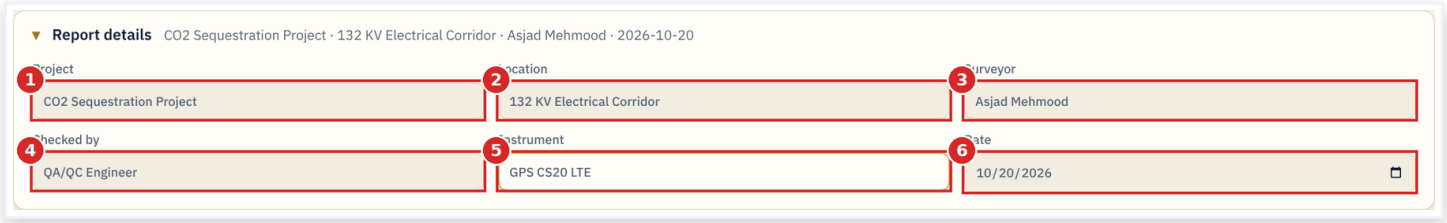
3 1 Known points
Station, backsight and closing point.

4 2 Site coordinates
Points as measured, in order observed.

5 Result panel
Updates live as you type.

TIP Your entries are saved on this device as you type, so closing the app does not lose work.

Report details



1. **Project**
   Printed at the top of every PDF report.

2. **Location**
   Site or corridor name.

3. **Surveyor**
   Your account name. Edit it if needed.

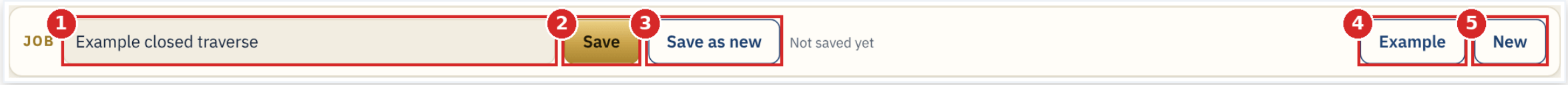
4. **Checked by**
   Checker or QA/QC engineer.

5. **Instrument**
   Total station or GPS model.

6. **Date**
   Today by default. Change with the calendar button.

TIP Click the "Report details" bar to open or close it. The same details are used on traverse and levelling reports.

Job bar: name, save, example, new



1 Job name

Type a name such as "Al Khor TP loop 1". If you leave it empty, one is made from the project and date.

2 Save

Stores the job under My jobs on the Dashboard. The text beside it shows Not saved yet, Unsaved changes or Saved.

3 Save as new

Makes a separate copy with "(copy)" added, so you can try a variation.

4 Example

Loads sample data for the traverse type you have selected, closed or open.

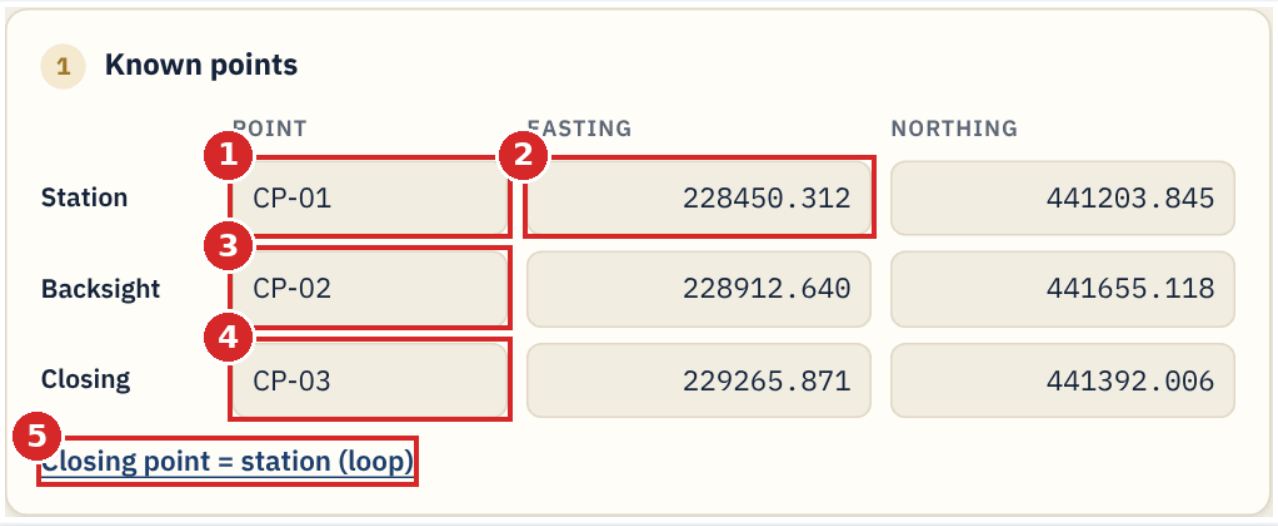
5 New

Clears the screen after you confirm with Start new. Unsaved changes are lost.



Traverse type switch. Choose Closed traverse or Open traverse. Your known points and readings stay when you switch.

Step 1: Known points



1 Station name
The known point where the traverse starts, for example CP-01.

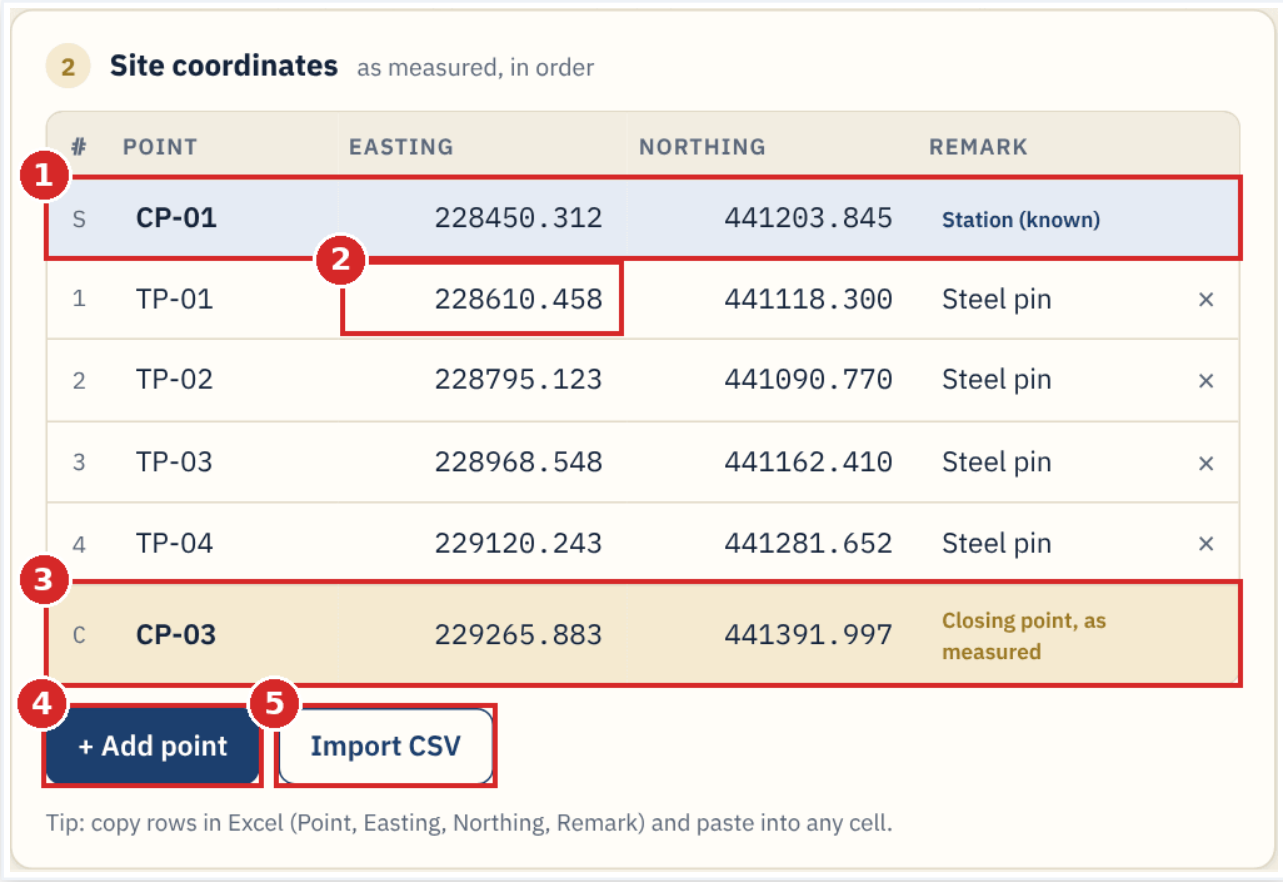
2 Station Easting and Northing
Known coordinates in metres. Commas and spaces are accepted. A box turns red if it is not a number.

3 Backsight name
A second known point that gives the starting direction. It must not have the same coordinates as the station.

4 Closing name
The known point where the traverse ends. It is used for the closure check.

5 Closing point = station (loop)
One click copies the station into the closing point for a loop that returns to the start.

Step 2: Site coordinates



1. **Row S: station**
   Known station coordinates from step 1. Not editable here.

2. **Traverse rows**
   One row per point, in order. The x deletes a row.

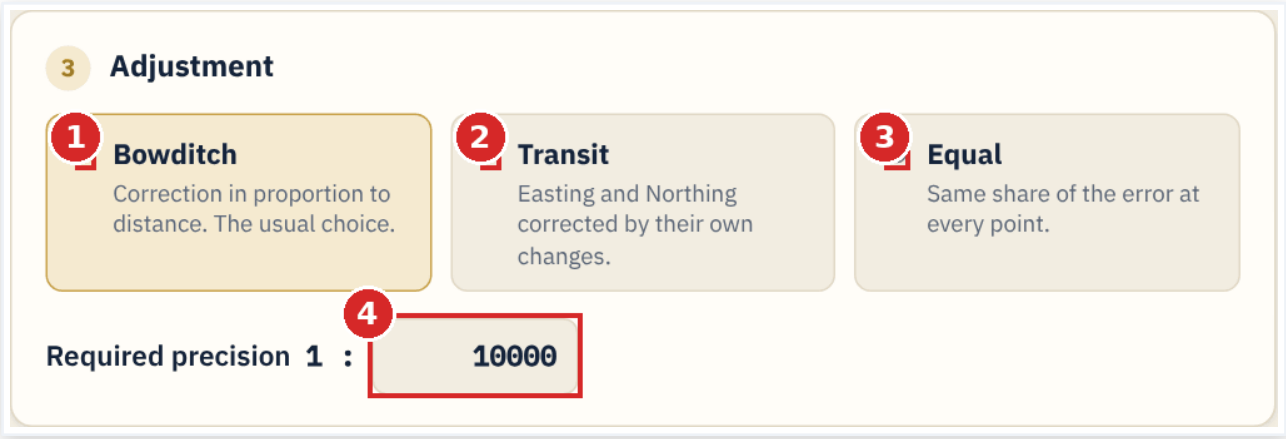
3. **Row C: closing point**
   Coordinates as measured on site, not the known values.

4. **+ Add point**
   Adds an empty row at the bottom.

5. **Import CSV**
   Loads Point, Easting, Northing, Remark.

TIP Copy rows in Excel (Point, Easting, Northing, Remark) and paste into any cell. The rows fill automatically.

Step 3: Adjustment method and precision



1 Bowditch (compass rule)
The usual choice. Each correction is in proportion to the distance travelled.

2 Transit
Easting and Northing are corrected in proportion to their own changes. It needs some change in both directions.

3 Equal
Every point receives the same share of the error.

4 Required precision 1 : N
Type N only, for example 10000 for 1:10,000. The result is accepted when the achieved precision is equal or better.

TIP Use the method your specification asks for. The default is Bowditch.

Reading the result



1. **Status banner**
   Green Accepted when the achieved precision is equal to or better than required, for example 1 : 62,105 against 1 : 10,000.

2. **Misclosure tiles**
   Misclosure E and N and the linear misclosure in millimetres, the precision achieved, and the total traverse length.

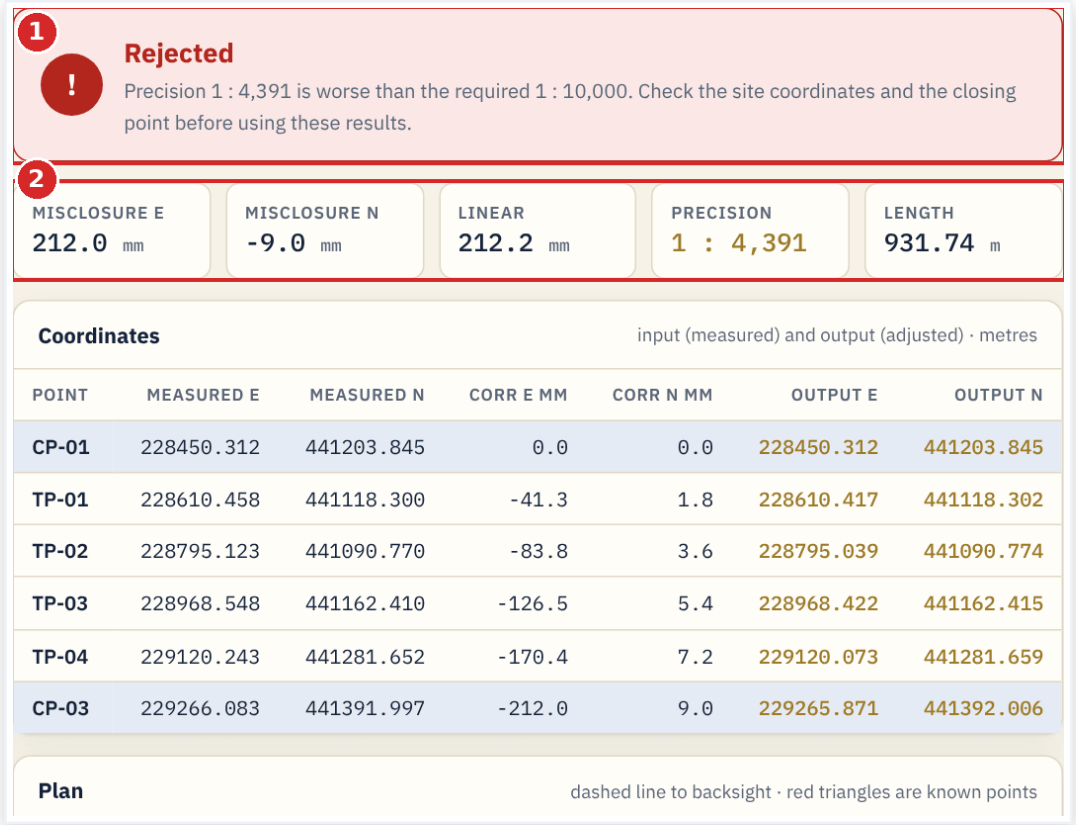
3. **Coordinates table**
   Measured E and N, the corrections in millimetres, and the Output E and N. Use the Output (adjusted) columns.

4. **Plan sketch**
   Gold line is adjusted, dashed line is as measured, red triangles are known points, dotted line goes to the backsight.

5. **PDF report**
   Creates the signed-off report for the job.

6. **Coordinates CSV**
   Exports the table for Excel or CAD.

When the result is rejected

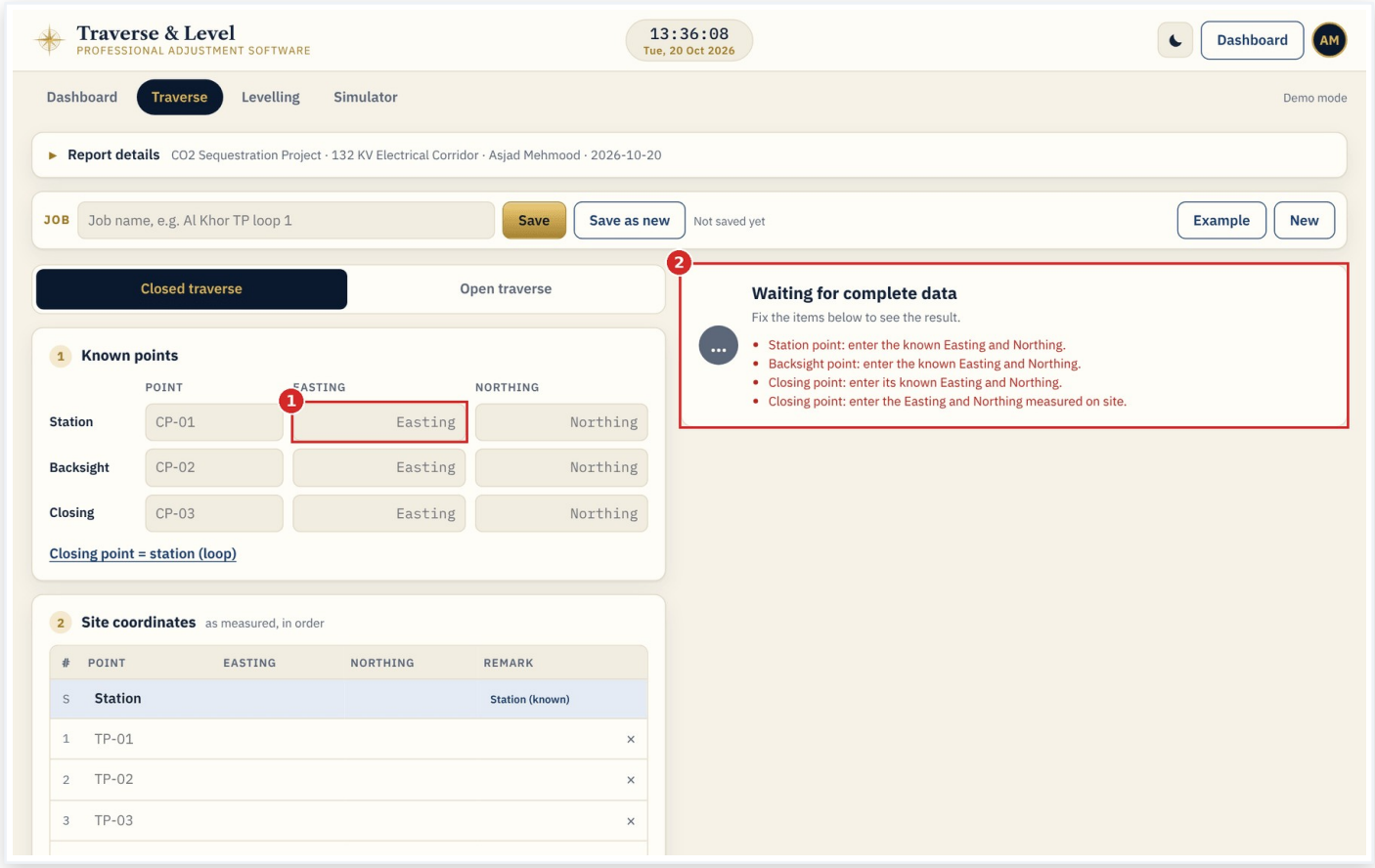


1. **Red Rejected banner**
   The achieved precision is worse than required. Here 1 : 4,391 against 1 : 10,000.

2. **Misclosure tiles**
   Show how large the error is. Linear misclosure 212.0 mm on 931.74 m is too large for 1 : 10,000.

NOTE Do not use rejected results. Check that known and measured closing values are not swapped, re-check typed coordinates, then re-observe if needed.

Waiting for complete data

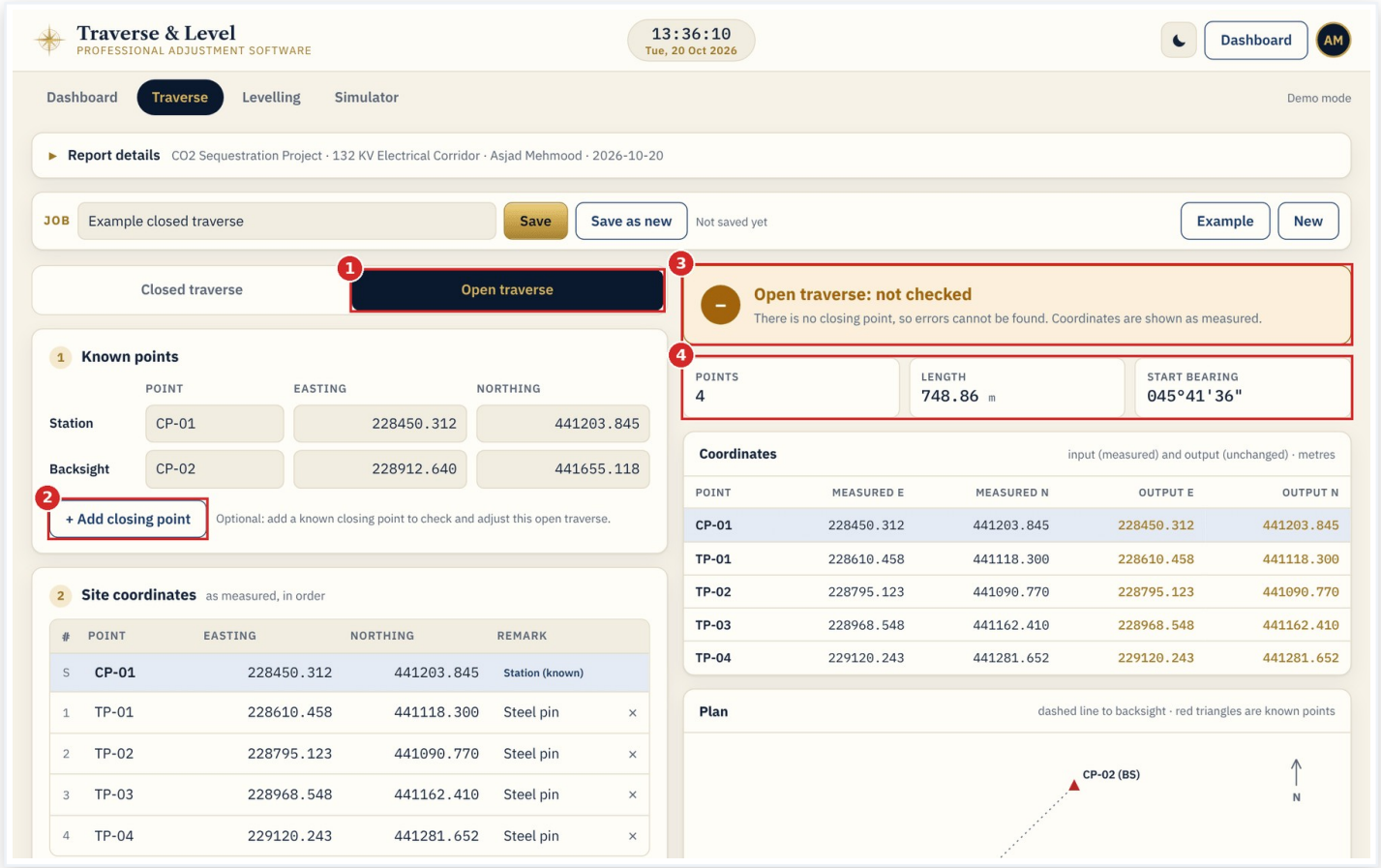


1 **Empty or invalid box**
A box with a red outline is missing or is not a number.

2 **Status list**
Lists exactly what to fix, one line each. The result appears when the list is empty.

TIP Typical messages: enter known Easting and Northing; row N is missing a number; backsight has the same coordinates as the station; precision must be positive.

Open traverse: as measured



1 Open traverse
Select it to run a traverse that does not end on a known point.

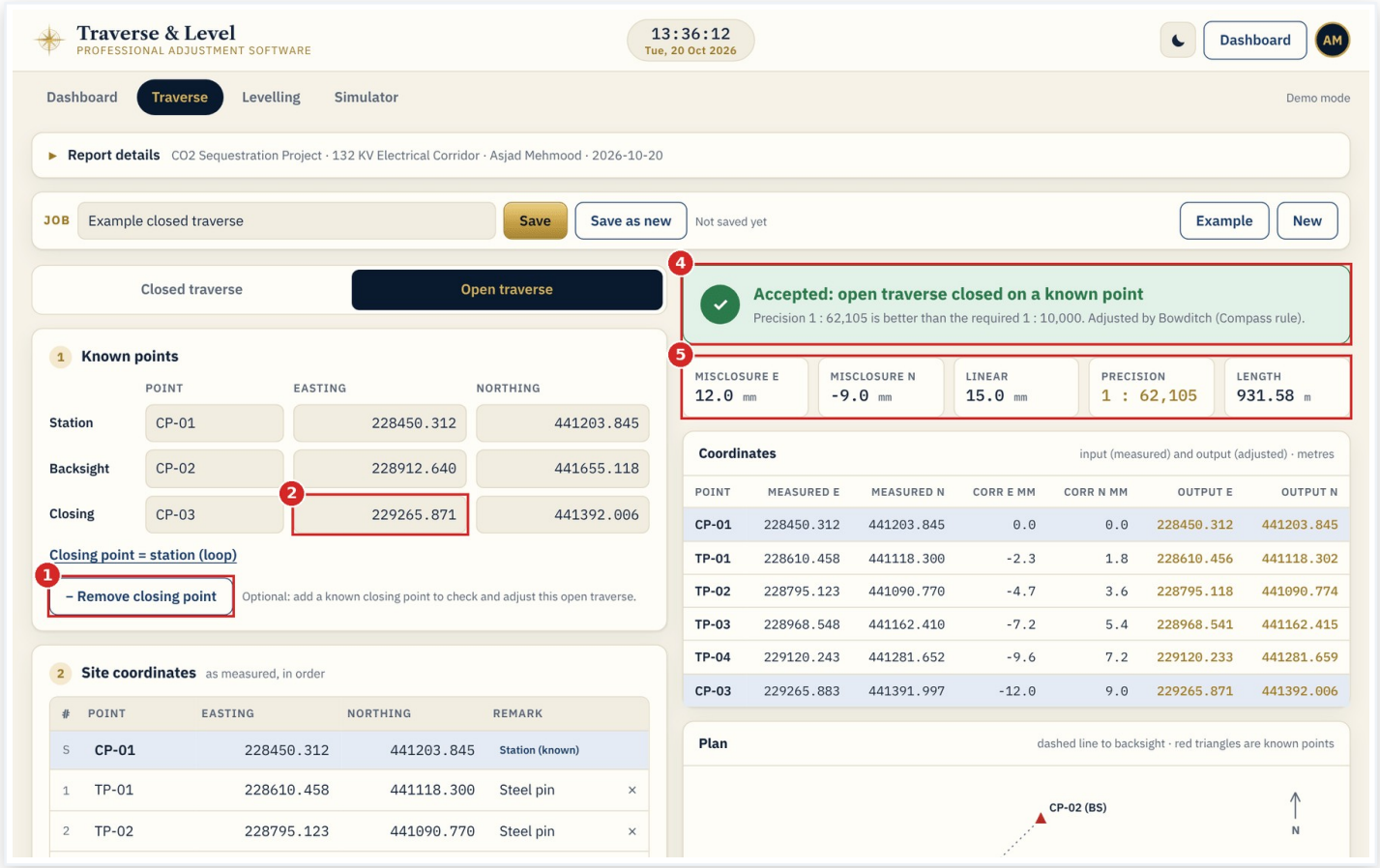
2 Add closing point
New button. It is optional and is explained on the next slide.

3 Not checked banner
Without a closing point errors cannot be found, so coordinates are shown exactly as measured.

4 Summary tiles
Number of points, length and the starting bearing.

TIP Choose Example while Open traverse is selected to load an open sample.

New: add a closing point



1 + Add closing point
Press it under Known points. The button then reads Remove closing point.

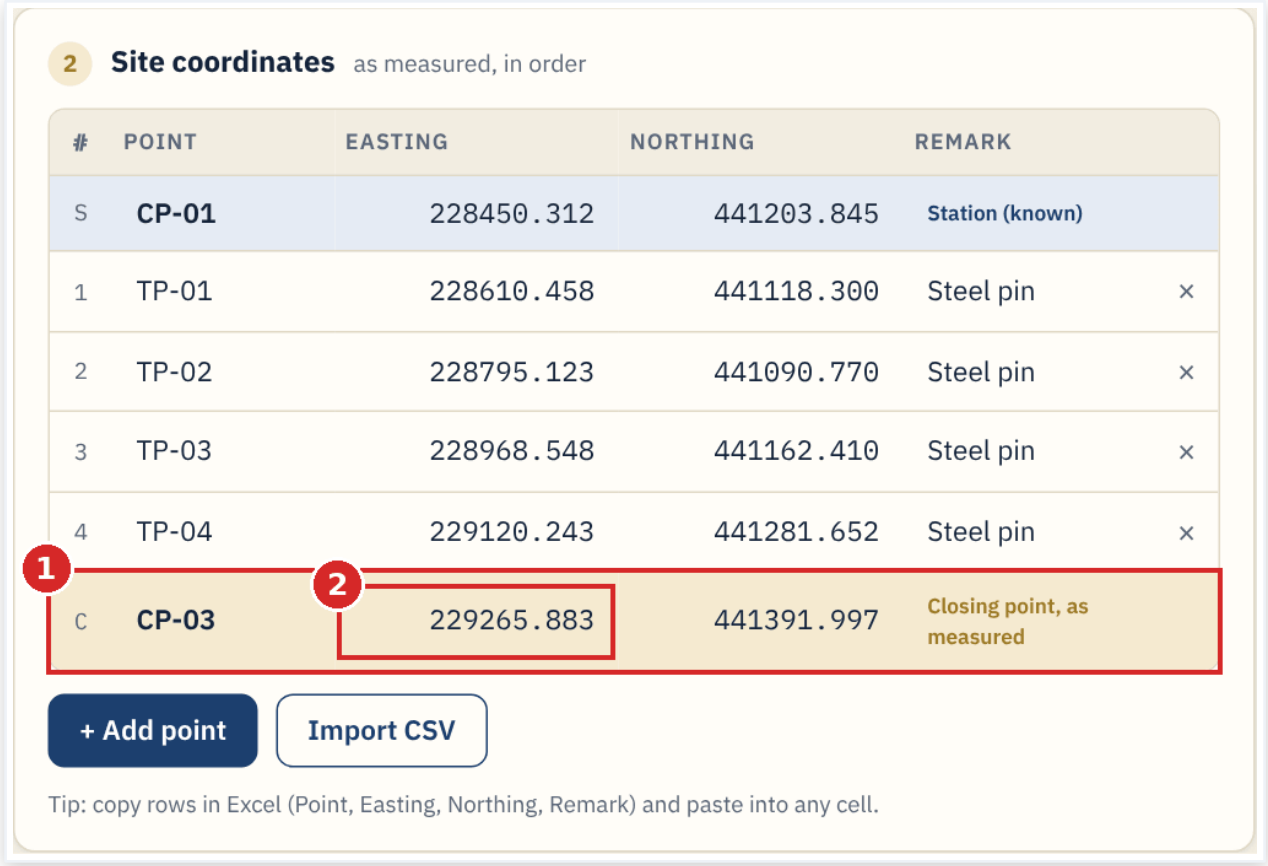
2 Closing point known coordinates
Type the name, Easting and Northing of the known point you finish on.

3 Adjustment card
Appears now: choose Bowditch, Transit or Equal and the required precision.

4 Status banner
Reads "Accepted: open traverse closed on a known point" or Rejected.

5 Misclosure tiles
Misclosure E and N, linear misclosure, precision and length.

Enter the measured closing point



1. **Row C appears**
   A new row at the bottom of Site coordinates, tagged "Closing point, as measured".

2. **Measured Easting and Northing**
   Type what you observed on site for the closing point. The known values stay in Known points.

NOTE Both the known and the measured closing coordinates are needed. The status shows what is missing until both are complete.

Open traverse result



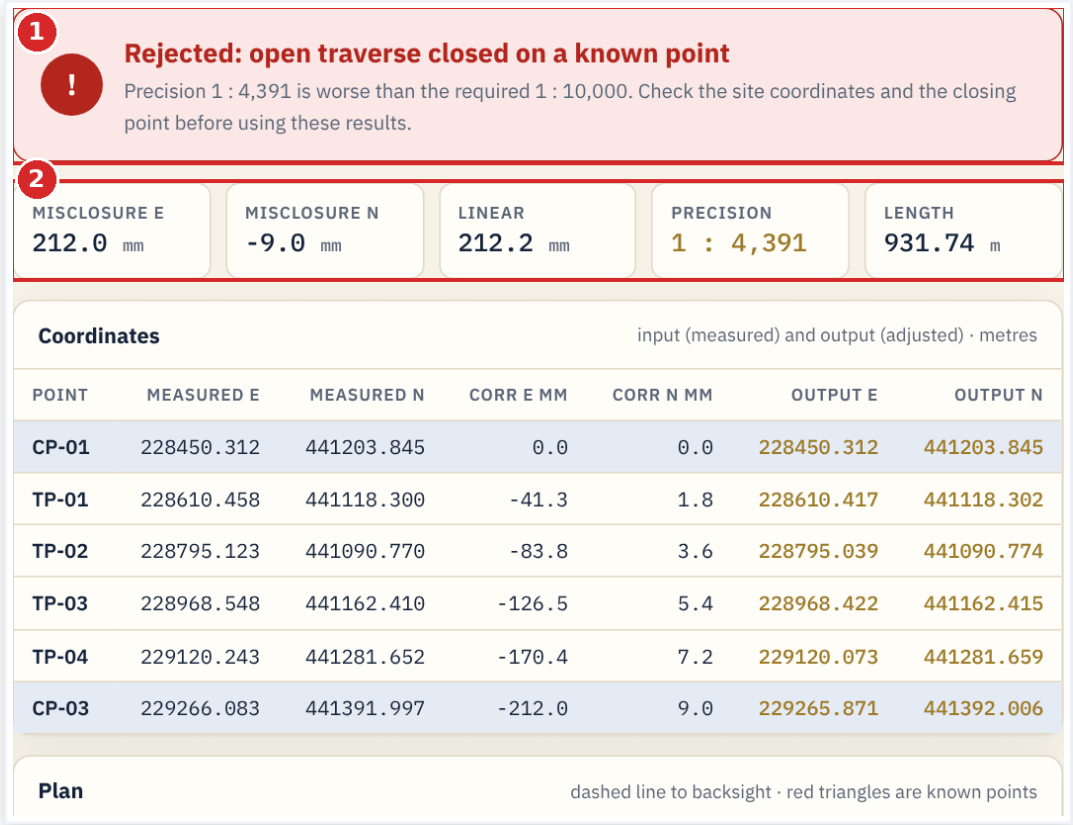
1 Accepted banner
Precision 1 : 62,105 is better than the required 1 : 10,000, adjusted by Bowditch.

2 Misclosure tiles
12.0 mm East and -9.0 mm North, linear 15.0 mm, over 931.58 m.

3 Coordinates table
Corrections grow along the traverse. The closing point ends on its known coordinates.

4 Plan sketch
Dashed line shows as measured and the gold line adjusted. The red triangle at the end is the known closing point.

Rejected, or remove the closing point

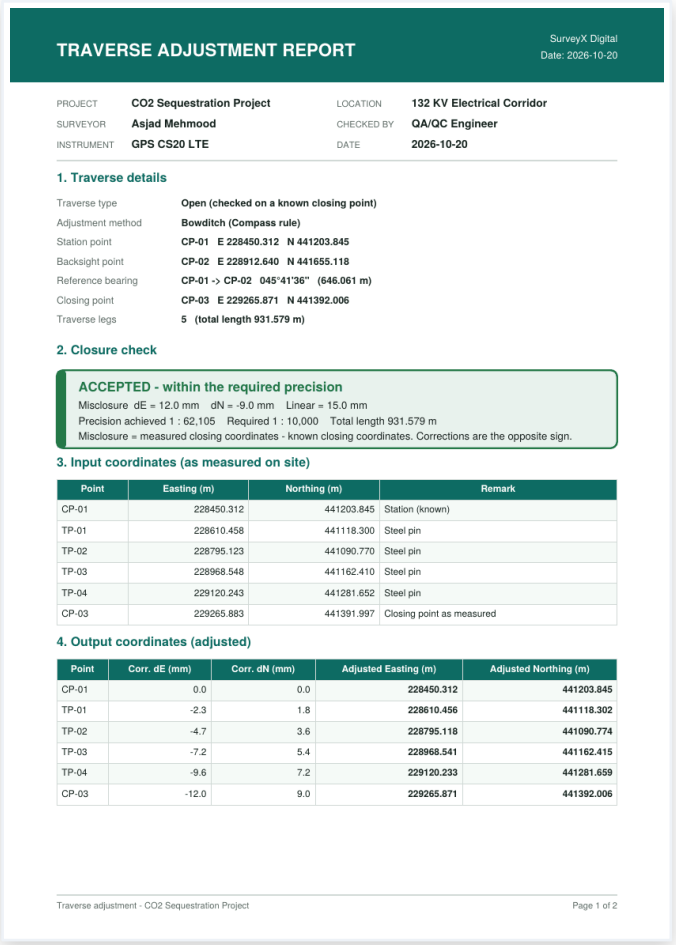


1 **Rejected: open traverse closed on a known point**
The misclosure is too large for the required precision. Check the closing coordinates and re-observe if needed.

2 **Remove closing point**
Press the button again to go back to the unchecked open traverse. Your typed values are kept.

NEW Closed traverse and open traverse without a closing point behave exactly as before.

PDF report and CSV export



TRAVERSE ADJUSTMENT REPORT

SurveyX Digital
Date: 2026-10-20

PROJECT	CO2 Sequestration Project	LOCATION	132 KV Electrical Corridor
SURVEYOR	Asjad Mehmood	CHECKED BY	QA/QC Engineer
INSTRUMENT	GPS CS20 LTE	DATE	2026-10-20

1. Traverse details

Traverse type	Open (checked on a known closing point)
Adjustment method	Bowditch (Compass rule)
Station point	CP-01 E 228450.312 N 441203.845
Backsight point	CP-02 E 228912.640 N 441655.118
Reference bearing	CP-01 -> CP-02 045°41'36" (646.061 m)
Closing point	CP-03 E 229265.871 N 441392.006
Traverse legs	5 (total length 931.579 m)

2. Closure check

ACCEPTED - within the required precision
Misclosure dE = 12.0 mm dN = -9.0 mm Linear = 15.0 mm
Precision achieved 1 : 62,105 Required 1 : 10,000 Total length 931.579 m
Misclosure = measured closing coordinates - known closing coordinates. Corrections are the opposite sign.

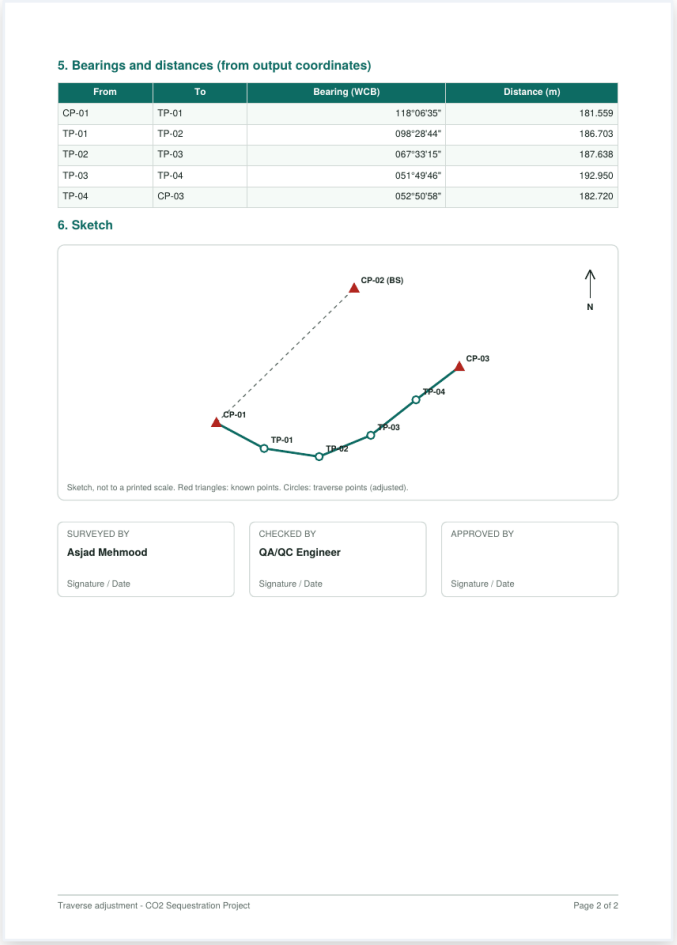
3. Input coordinates (as measured on site)

Point	Easting (m)	Northing (m)	Remark
CP-01	228450.312	441203.845	Station (known)
TP-01	228610.458	441118.300	Steel pin
TP-02	228795.123	441090.770	Steel pin
TP-03	228968.548	441162.410	Steel pin
TP-04	229120.243	441281.652	Steel pin
CP-03	229265.883	441391.997	Closing point as measured

4. Output coordinates (adjusted)

Point	Corr. dE (mm)	Corr. dN (mm)	Adjusted Easting (m)	Adjusted Northing (m)
CP-01	0.0	0.0	228450.312	441203.845
TP-01	-2.3	1.8	228610.456	441118.302
TP-02	-4.7	3.6	228795.118	441090.774
TP-03	-7.2	5.4	228968.541	441162.415
TP-04	-9.6	7.2	229120.233	441281.659
CP-03	-12.0	9.0	229265.871	441392.006

Traverse adjustment - CO2 Sequestration Project Page 1 of 2



5. Bearings and distances (from output coordinates)

From	To	Bearing (WCB)	Distance (m)
CP-01	TP-01	118°06'35"	181.559
TP-01	TP-02	098°28'44"	186.703
TP-02	TP-03	067°33'15"	187.638
TP-03	TP-04	051°49'46"	192.950
TP-04	CP-03	052°50'58"	182.720

6. Sketch

Sketch, not to a printed scale. Red triangles: known points. Circles: traverse points (adjusted).

SURVEYED BY	CHECKED BY	APPROVED BY
Asjad Mehmood	QA/QC Engineer	
Signature / Date	Signature / Date	Signature / Date

Traverse adjustment - CO2 Sequestration Project Page 2 of 2

PDF report

Press PDF report under the results. It contains: 1 Traverse details, 2 Closure check, 3 Input coordinates, 4 Output coordinates, 5 Bearings and distances, 6 Sketch.

Type line in the report

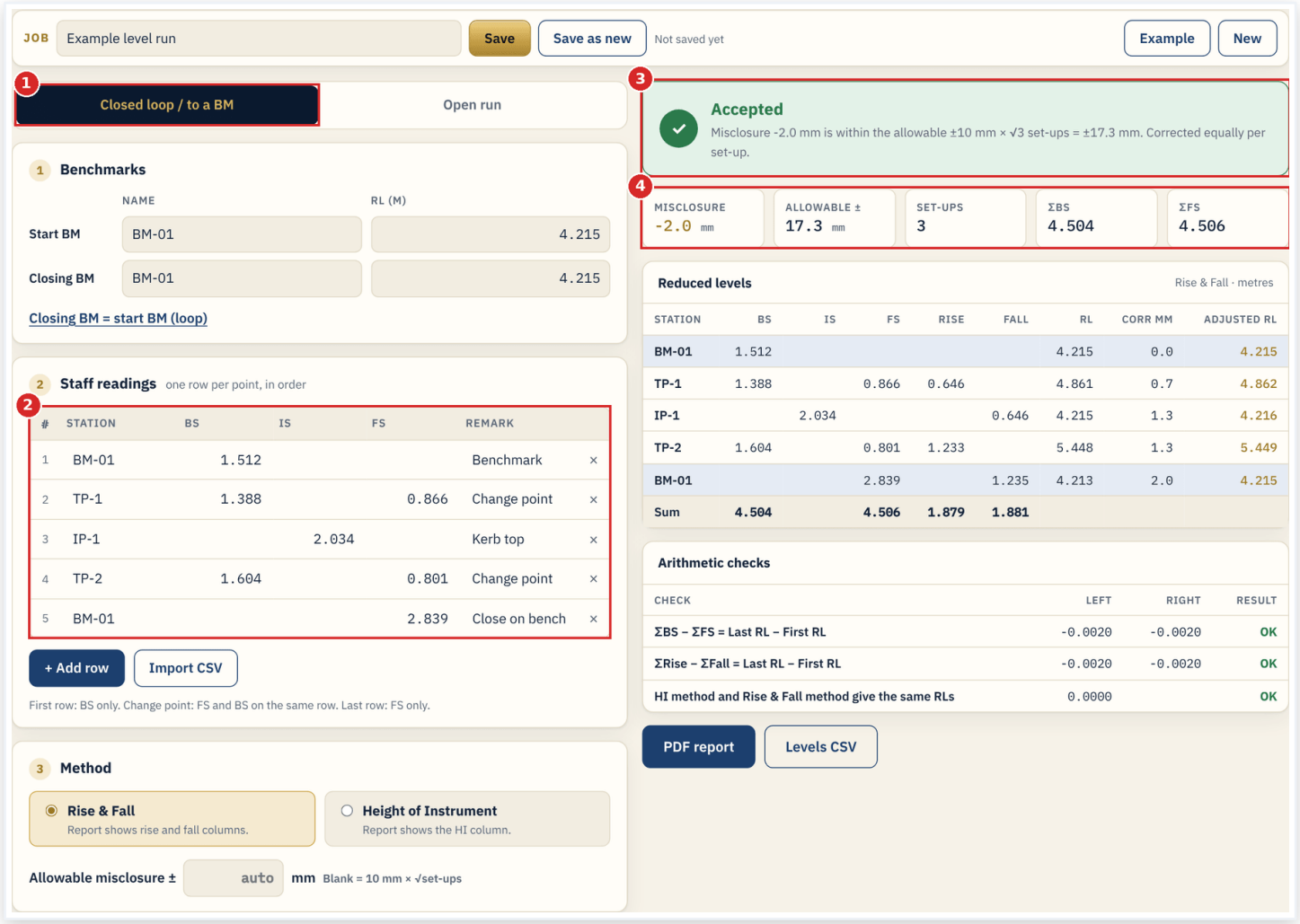
Reads Closed, Open (checked on a known closing point) or Open (no closing check).

Coordinates CSV

Press Coordinates CSV for Point, Measured E and N, Corr dE and dN, Output E and N, Remark.

TIP The PDF uses your Report details (project, location, surveyor, checker, instrument, date). Fill them in first.

The Levelling screen



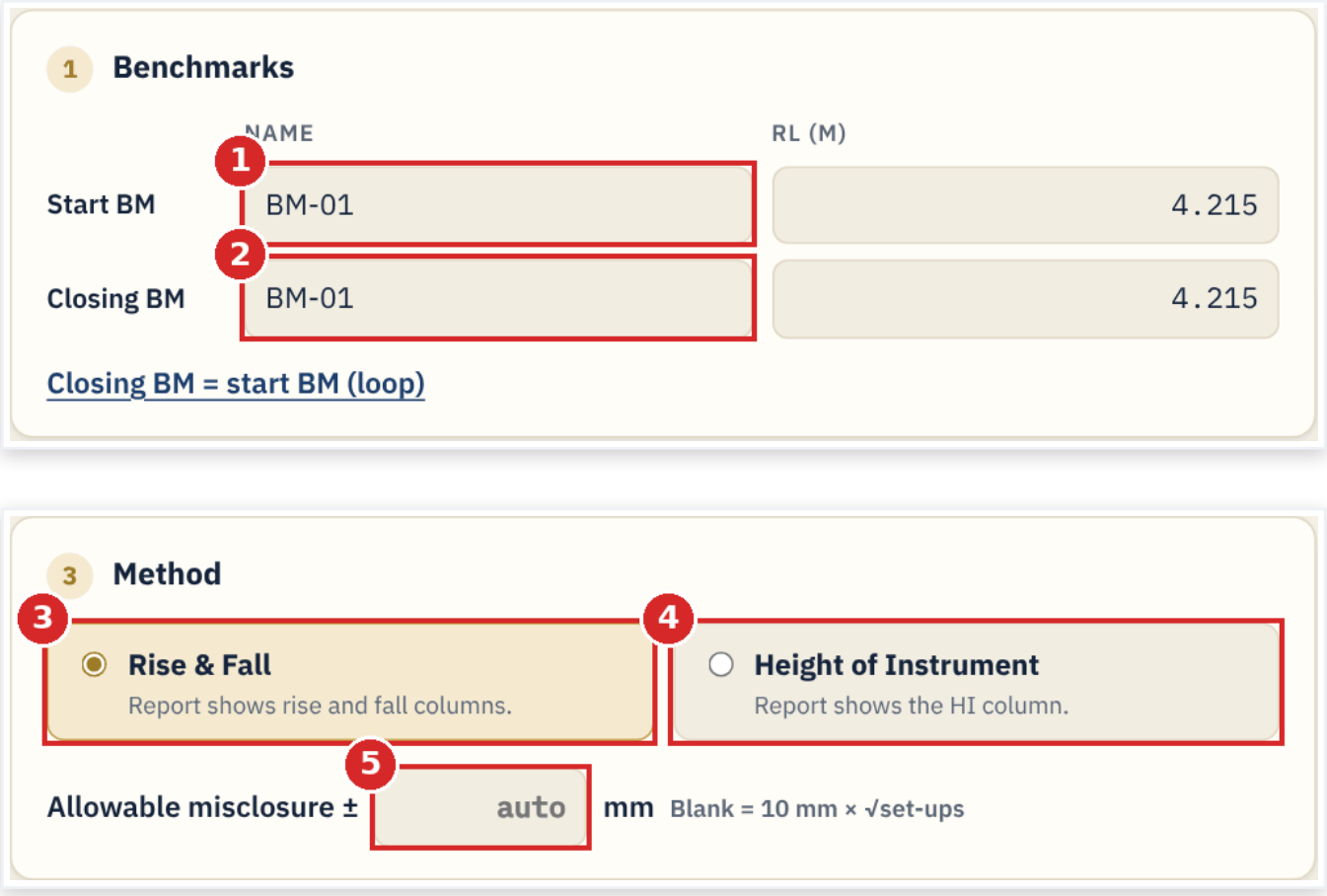
1 Run type
Closed loop / to a BM checks on a known benchmark. Open run is not checked.

2 Benchmarks and staff readings
Start BM with its RL, closing BM, then one row per staff reading.

3 Status banner
Accepted when the misclosure is within the allowable value, here -2.0 mm against 17.3 mm.

4 Summary tiles
Misclosure, allowable, number of set-ups, total BS and total FS.

Benchmarks and method



1. **Start BM**
   Name and known RL of the benchmark you start on.

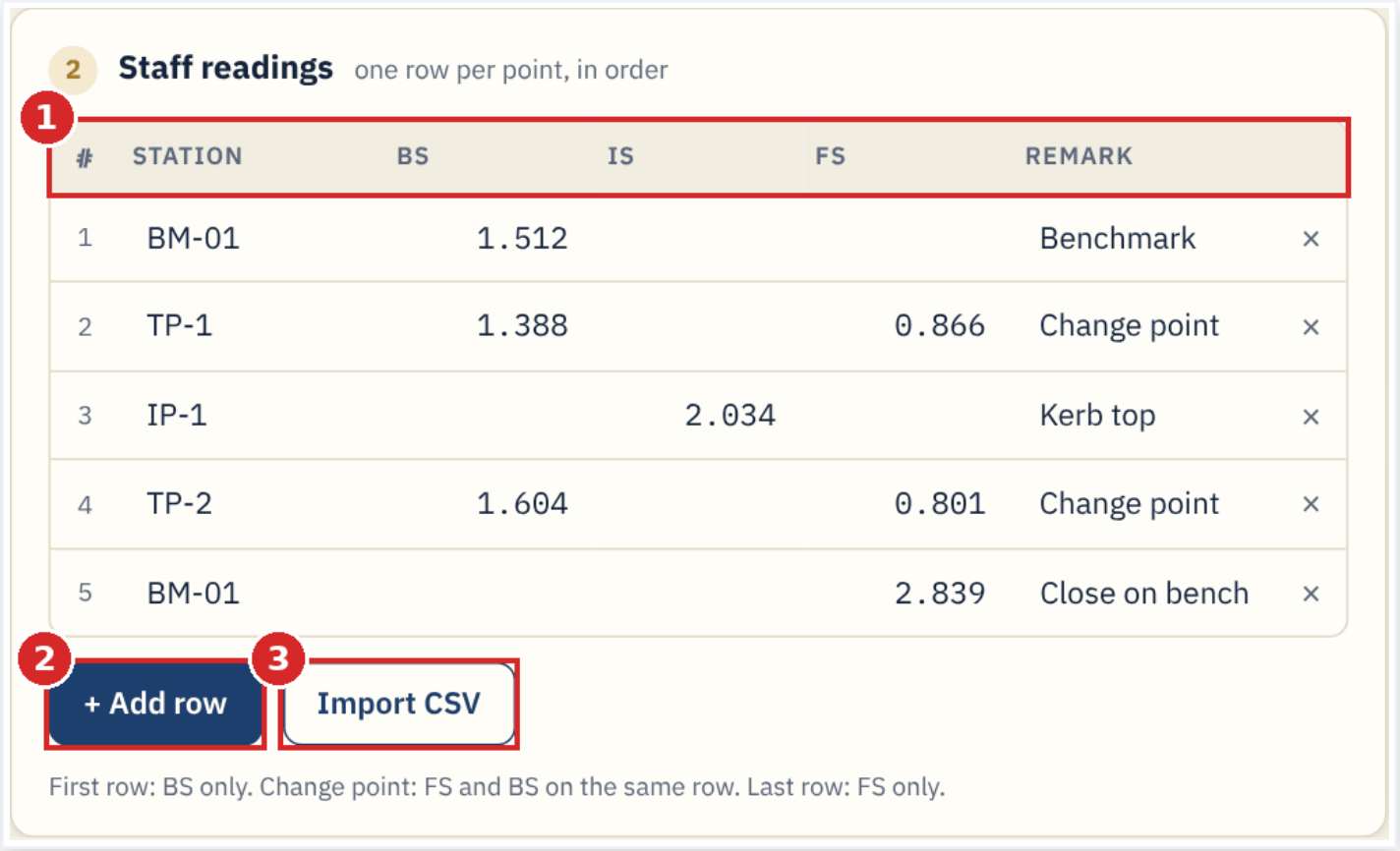
2. **Closing BM**
   Benchmark you finish on. For a loop press "Closing BM = start BM".

3. **Rise & Fall**
   Report shows rise and fall columns.

4. **Height of Instrument**
   Report shows the HI column. Both methods give the same levels.

5. **Allowable misclosure**
   Leave blank for 10 mm x sqrt(set-ups), or type your own value in mm.

Staff readings



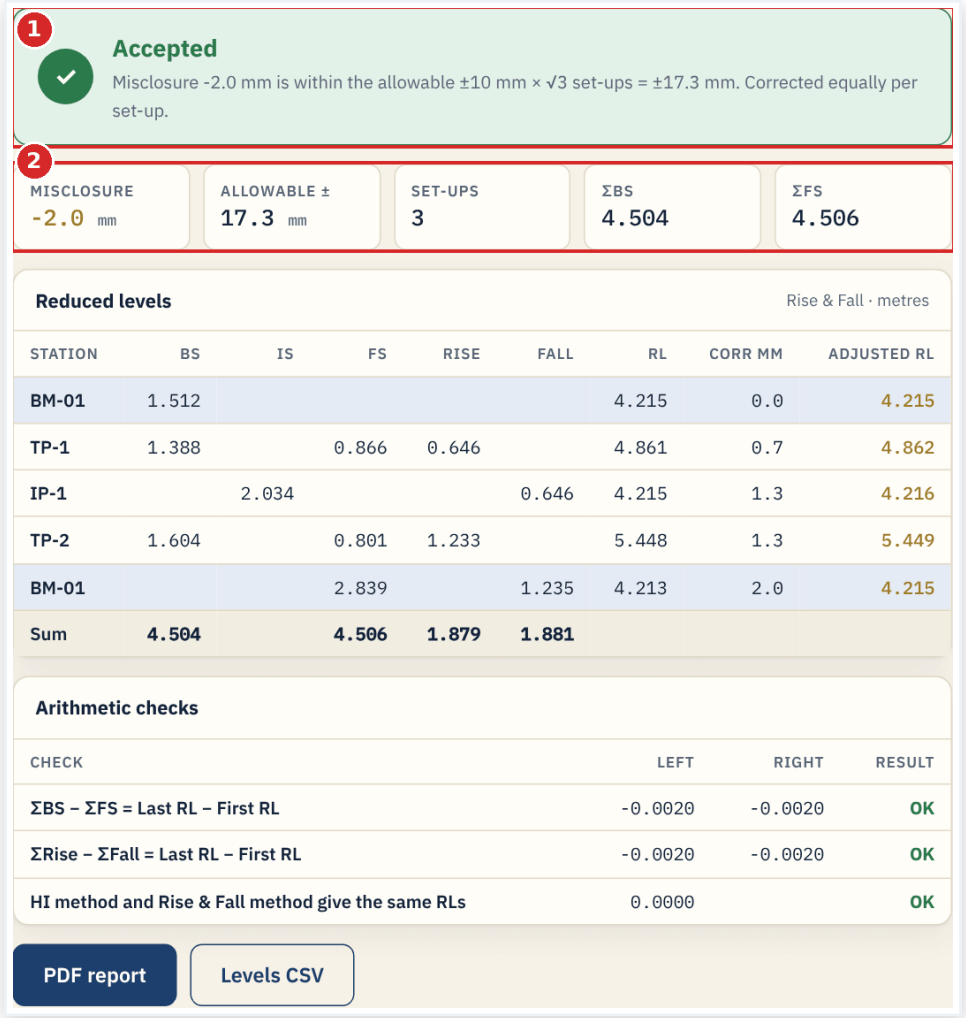
1 Readings table
One row per reading: Station, BS, IS, FS, Remark.

2 + Add row
Adds an empty row.

3 Import CSV
Loads Station, BS, IS, FS, Remark from a file.

TIP First row: BS only. Change point: FS and BS on the same row. Intermediate sight: IS alone. Last row: FS only.

Reading the levelling result



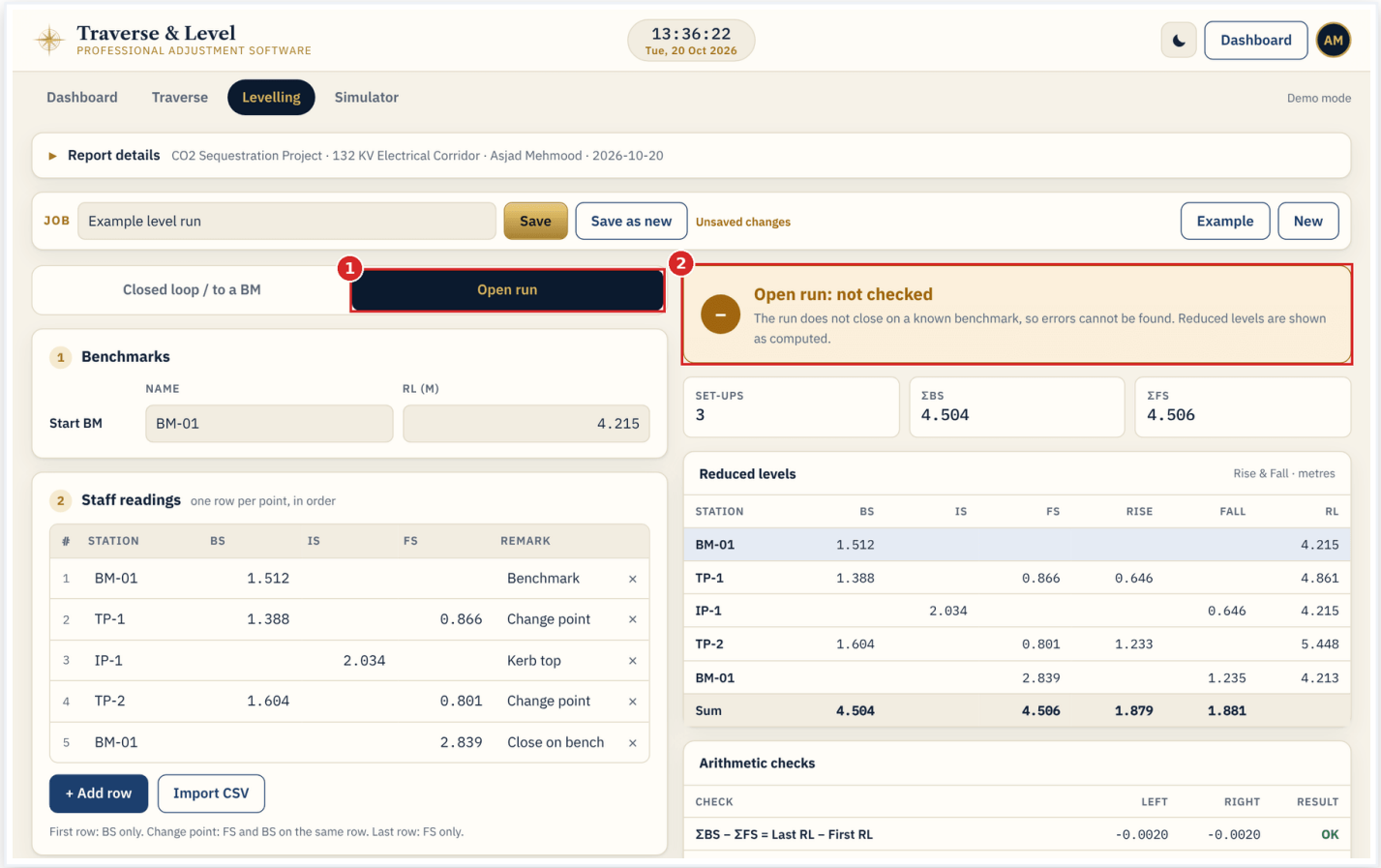
1. **Status and tiles**
   Accepted or Rejected, with the misclosure, allowable value, set-ups and sums of BS and FS.

2. **Reduced levels table**
   BS, IS, FS, Rise, Fall, RL, correction in mm and the adjusted RL. Correction is shared equally per set-up.

3. **Arithmetic checks**
   Sum BS - Sum FS and Sum Rise - Sum Fall must equal last RL - first RL. All checks show OK.

4. **PDF report and Levels CSV**
   Buttons at the bottom of the result column.

Open run

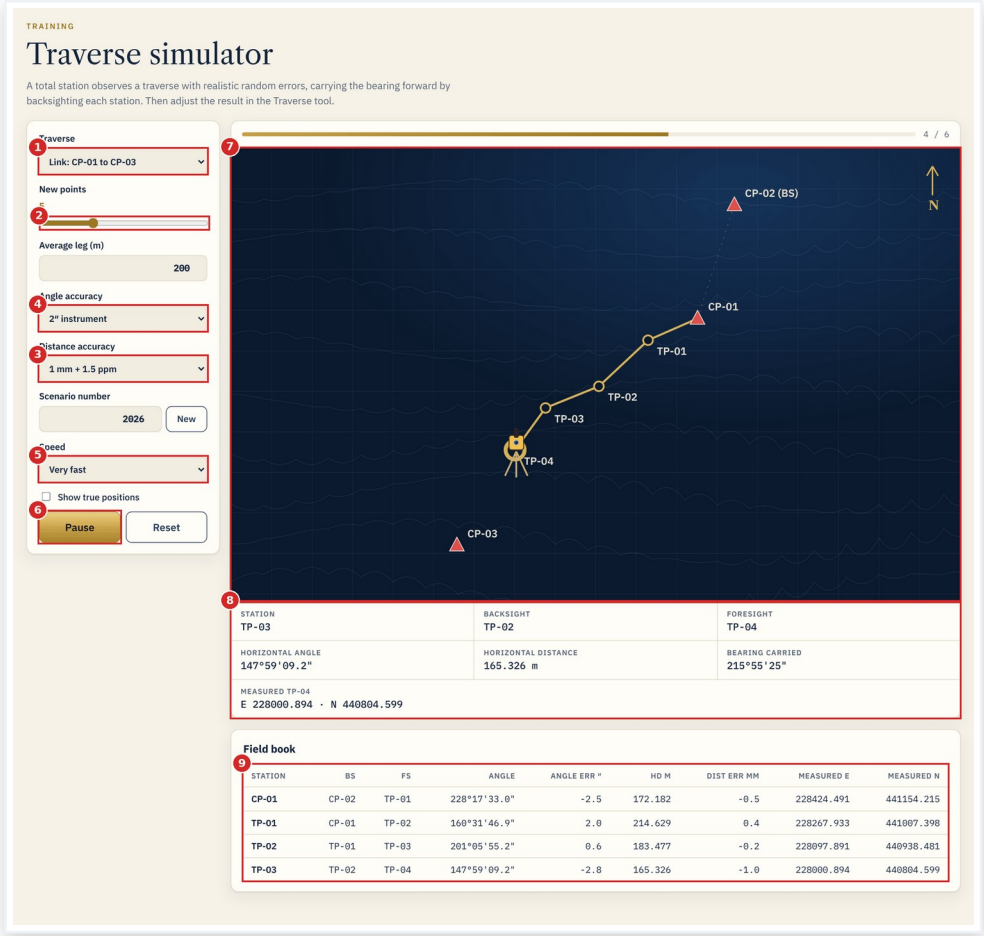


1 **Open run**
Select it when you do not close on a benchmark.

2 **Not checked banner**
Errors cannot be found. Reduced levels are shown as computed and no correction is applied.

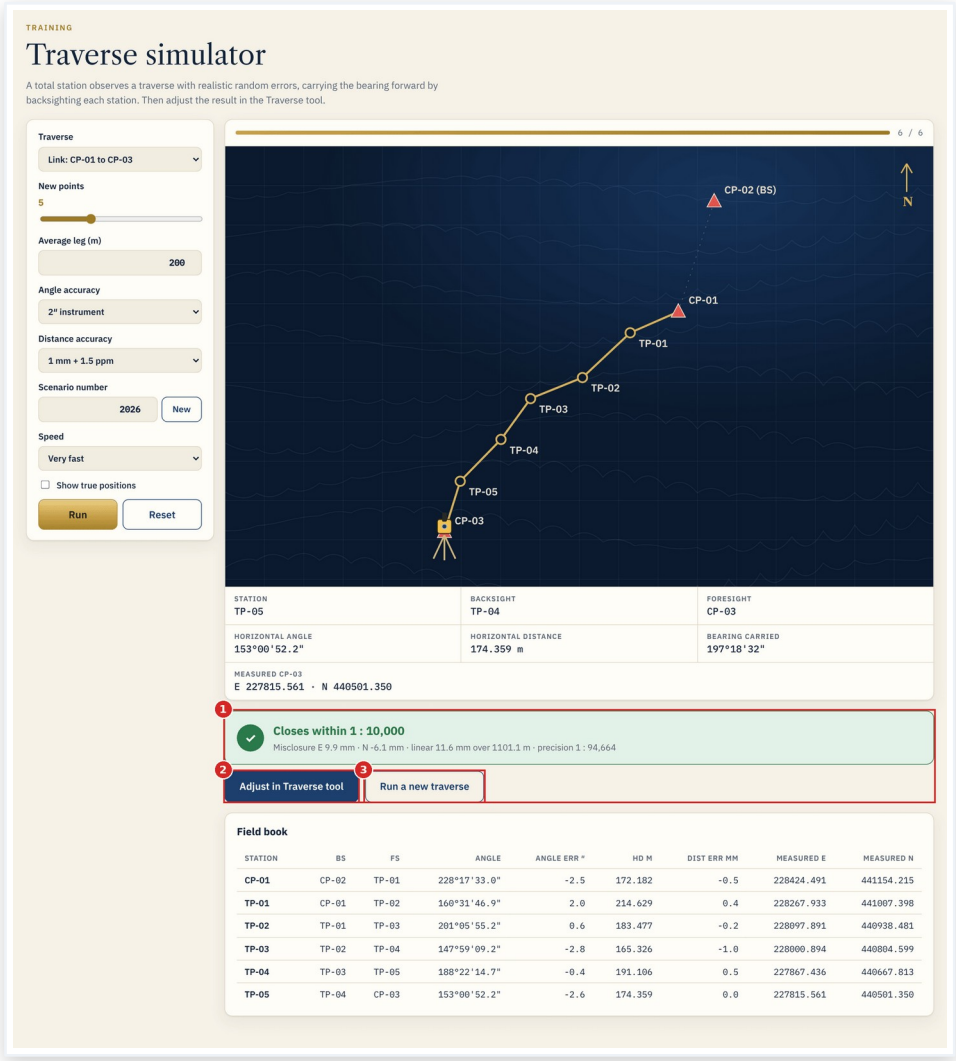
TIP The Closing BM boxes are hidden for an open run. Switch back to Closed loop to use them.

Traverse simulator



1. **Traverse** Link CP-01 to CP-03, or loop back.
2. **New points** Slider for the number of points.
3. **Distance accuracy** For example 1 mm + 1.5 ppm.
4. **Angle accuracy** 1 second up to a poor 30 second set-up.
5. **Speed** Slow to very fast.
6. **Run / Pause** Starts or pauses. Reset clears.
7. **Stage** Plan view with the moving instrument.
8. **Current reading** Station, backsight, angle, distance.
9. **Field book** Each reading with its errors.

Finish and adjust the simulated traverse



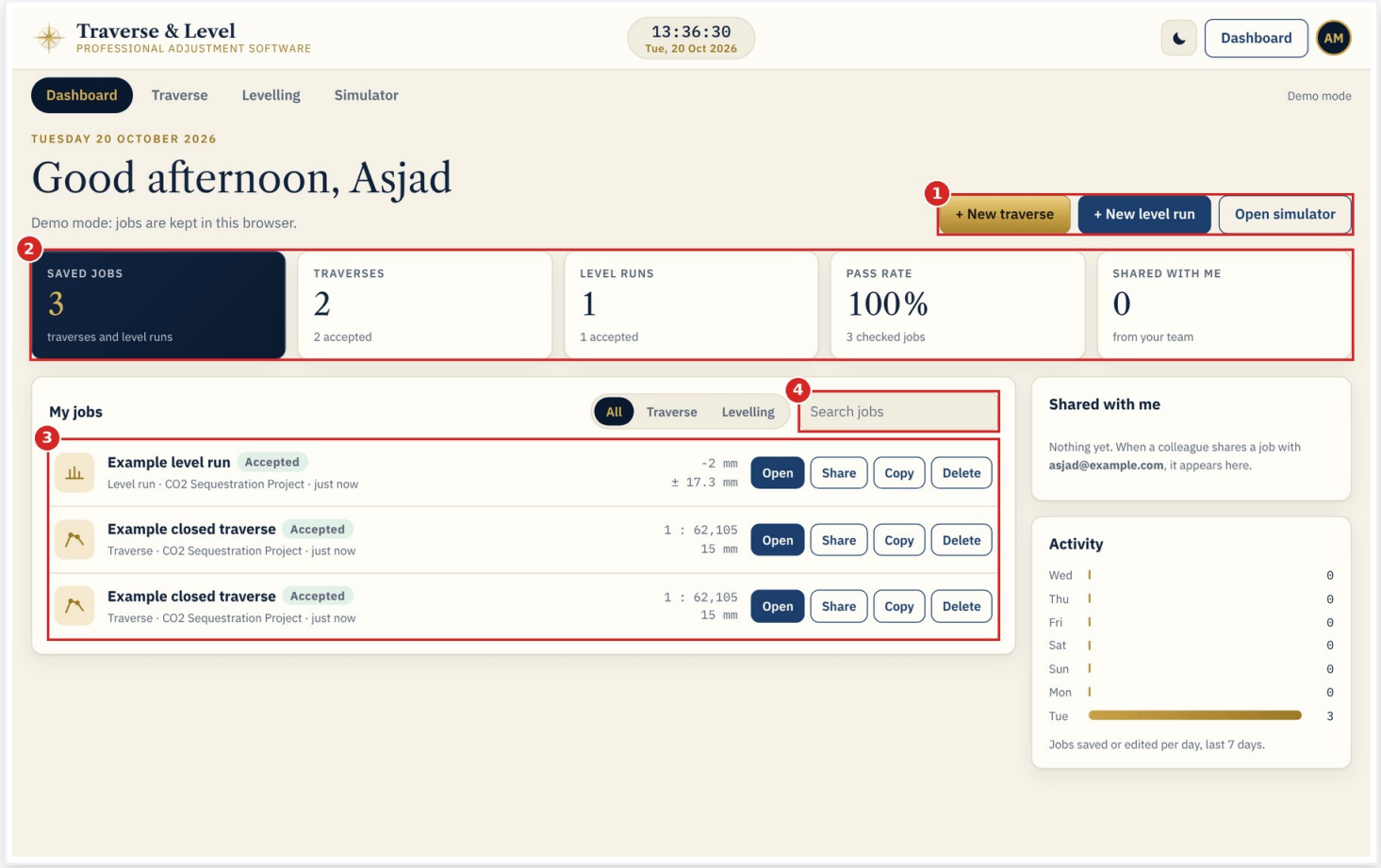
1 Result message
Tells how the simulated traverse closed, for example "Closes within 1 : 10,000" with the misclosure.

2 Adjust in Traverse tool
Sends the simulated points to the Traverse screen so you can adjust them.

3 Run a new traverse
Starts again with a new scenario number.

TIP Use the simulator for training. Try poor angle accuracy to see a traverse that gets rejected.

Dashboard and saved jobs



1 Quick buttons
Start a new traverse, a new level run, or open the simulator.

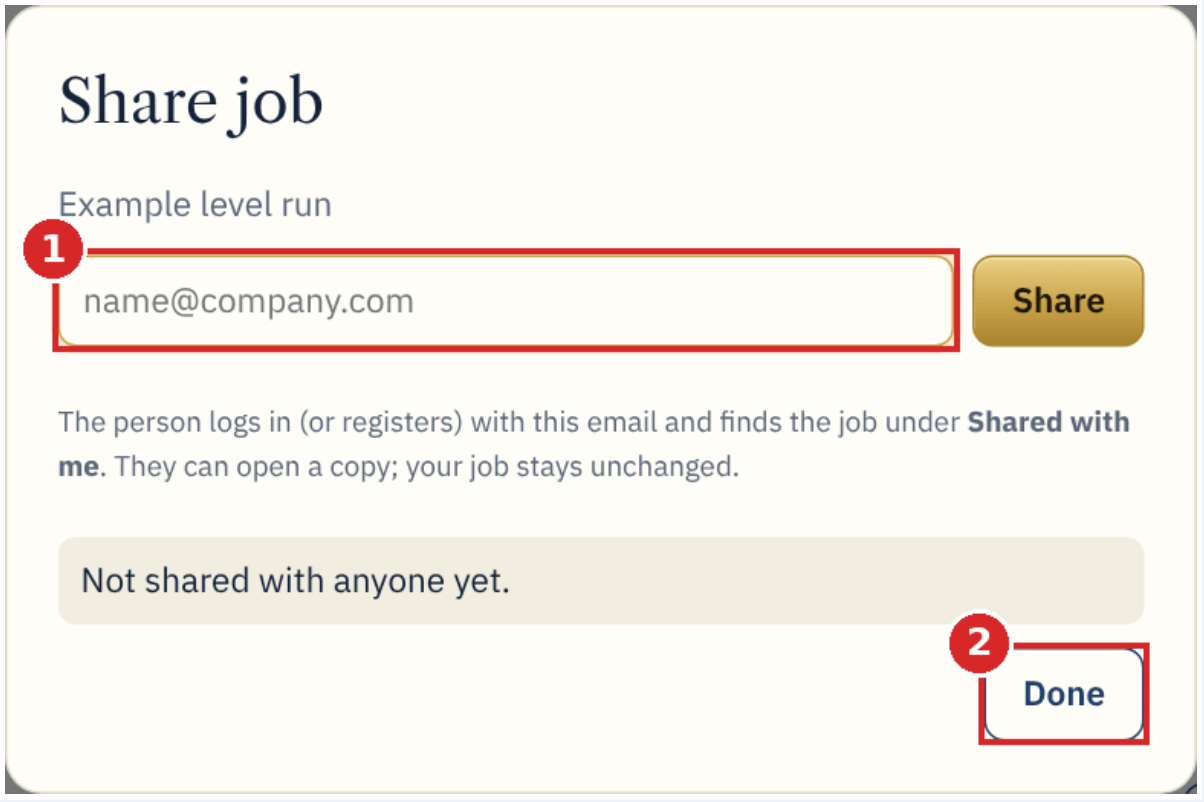
2 Summary tiles
Saved jobs, traverses, level runs, pass rate and jobs shared with you.

3 My jobs
Filter All, Traverse or Levelling. Each job has Open, Share, Copy and Delete.

4 Search jobs
Type part of a name to find a job.

TIP Shared with me lists jobs from colleagues. Activity shows jobs saved or edited per day over the last 7 days.

Share a job

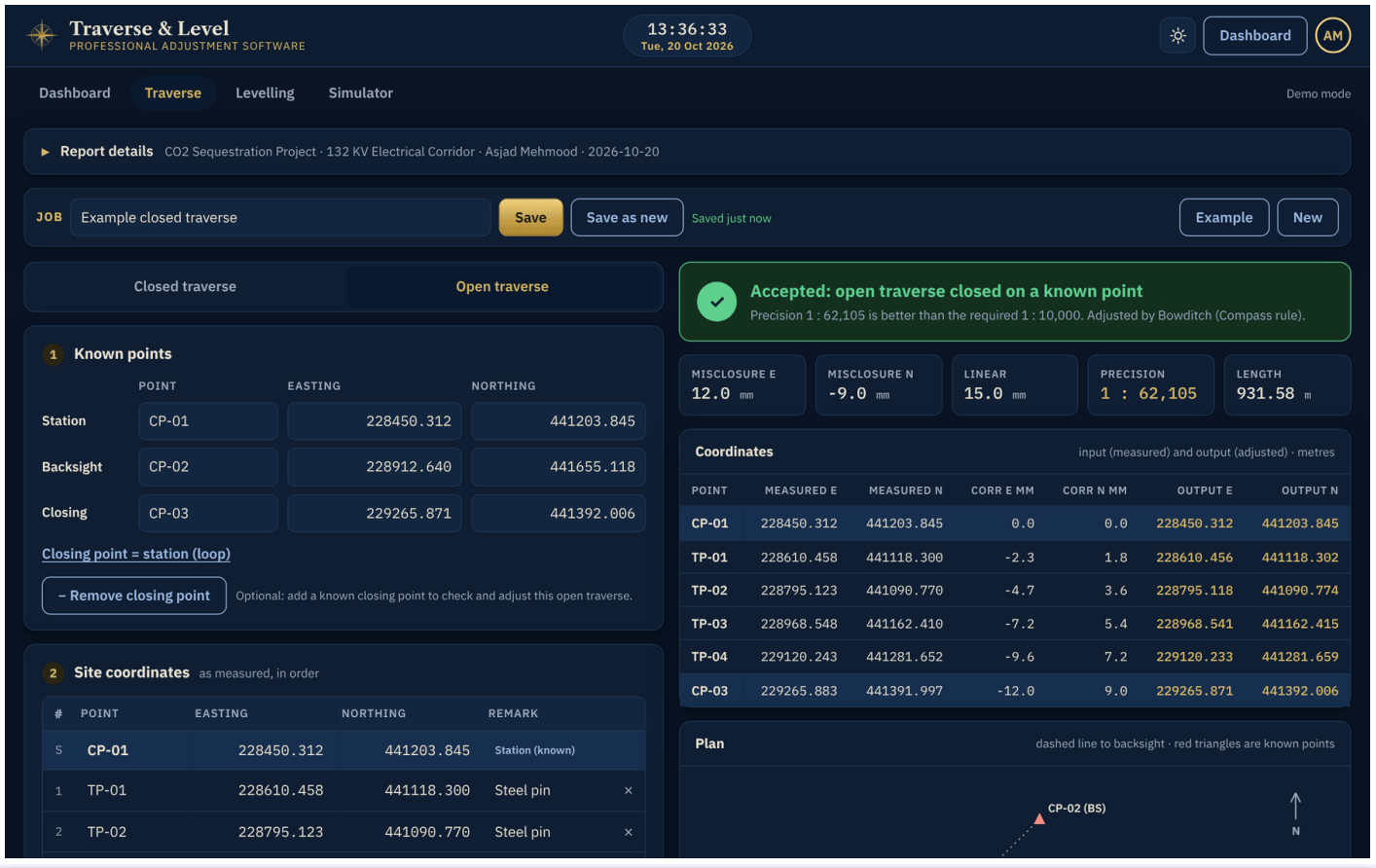


1. **E-mail address**
   Type the colleague's e-mail and press Share. They find the job under Shared with me.

2. **Done**
   Closes the dialog. The people you shared with are listed in it.

TIP The person can open a copy. Your job stays unchanged.

Dark mode and good practice



Dark mode
Use the moon button in the top bar. It suits low light and is remembered.

Save often
Press Save after the first complete result and again after edits.

Check inputs first
Known and measured closing values are the most common mix-up.

Report details
Fill them before the PDF so it is ready to issue.

Keep the original data
Use Save as new before trying another method.

QUICK REFERENCE

Which mode should I use?

	Closed traverse	Open traverse	Open + closing point (new)
Ends on	A known point	No known point	A known point
Closing point needed	Yes	No	Yes: known and measured
Misclosure and precision	Shown	Not shown	Shown
Adjustment	Bowditch, Transit or Equal	None	Bowditch, Transit or Equal
Status banner	Accepted or Rejected	Not checked	Accepted or Rejected
Output coordinates	Adjusted	As measured	Adjusted

Levelling follows the same idea: a closed loop or run to a BM is checked and adjusted, an open run is not checked.